Well-being, Aging in Place, and Use of the Kawa Model: A Pilot Study

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ABSTRACT

Introduction: Many older adults prefer to age in place, but are unprepared to do so because of lack of knowledge of strategies and available resources. Uncertainty and unpreparedness for upcoming life stages can detract from current psychosocial well-being as well as future success in aging in place.

Methods: This project explored the question of whether a Kawa model-based educational intervention could improve the current well-being of older adults related to aging in place. A pretest-posttest mixed methods design was used. Outcome measures included the Kawa activity, the Psychological General Well-Being Index, and an author-generated survey.

Results: Individualized education on aging in place improved psychosocial well-being in a small sample of community-dwelling older adults. The Kawa model was useful for eliciting client perceptions, establishing rapport, and enhancing communication.

Conclusion: Limitations of the methods used include a small, homogeneous sample (N = 7), a short time frame for measuring results, and relative inexperience of the first author with use of the Kawa model. Further studies are needed to explore the application of this model in occupational therapy interventions with older adults seeking to age in place. **[*Annals of International Occupational Therapy*. 2021;4(1);15-25.]**

The population of older adults is increasing more rapidly than that of other age demographics, both in the United States and throughout much of the world (Ortman et al., 2015; United Nations, 2013). In response, health care in the United States has been trending toward a more community-based, preventive model of care to manage the costs and logistics of caring for a growing population of elders. This model has been described in terms of a "triple aim" for health care: improving the experience of care, optimizing health outcomes, and increasing cost-effectiveness (Berwick et al., 2008). More recently, a fourth aspect has been suggested (evolving the concept to a "quadruple aim"), which is ensuring the well-being of health care providers and preventing burnout to more successfully achieve the original three goals (Bodenheimer & Sinsky, 2014).

The concept of aging in place has emerged within the past 35 years and aligns with this approach. Aging in place can be defined as "the ability to live in one's own home and community safely, independently, and comfortably," regardless of demographic factors, such as age or income level, and despite the changes to functional abilities that can occur during the aging process (Centers for Disease

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Control and Prevention, 2009). The movement toward aging in place seeks to provide a long-term, cost-effective solution to institutionalized health care by enabling older adults to remain in their own homes safely throughout their lives.

Aging in place has financial benefits. When their home environments, routines, and occupations are optimized for safety and independence, older adults need less professional care. As a result, aging in place is significantly more affordable than institutionalized care (U.S. Department of Health and Human Services, 2017).

Quality of life issues are additional factors when considering aging and aging in place. Planned or unplanned, relocation to a health care facility causes significant disruptions to the ability to participate in valued occupations, roles, habits, and routines. This ability is inextricably tied to well-being, identity, and quality of life (American Occupational Therapy Association, 2014). Abrupt role changes in older adulthood threaten place identity, cause relational discord, and increase psychological stress (Downey et al., 2017; O'Connell et al., 2010).

Although the overwhelming majority of U.S. older adults prefer and expect to age in place (Khalfani-Cox, 2017), recent studies have shown variation in their knowledge and perception of available resources and their preparation for aging in place (Johansson et al., 2009; Peek et al., 2016; Tang & Pickard, 2008). Older adults have shown varied levels of readiness to use compensatory strategies, such as environmental and behavioral modifications, to improve their chances of successfully aging in place (Rose et al., 2010). Their perceptions of their potential for aging in place and the available resources have also been linked to overall well-being (Ahn et al., 2020).

Studies indicate that there are important psychosocial benefits to the use of a proactive approach to aging in place. Early discussion of behavioral and environmental modifications as compensatory strategies can improve acceptance of and readiness for use of these strategies (Horowitz et al., 2013; Rose et al., 2010). Proactive preparation for aging in place and age-related changes improves older adults' current psychosocial well-being (Downey et al., 2017; O'Connell et al., 2010). Programs that provide more general preventive education and training in wellness, health promotion, and issues associated with aging have been shown to increase overall psychosocial well-being (Chua & de Guzman, 2014; Clark et al., 2012; Johansson & Björklund, 2015; O'Connell et al., 2010).

To date, no studies have specifically explored whether occupational therapy interventions to provide education and resources for aging in place have a direct effect on well-being. This gap in information provided the impetus for the research question: Does education on aging in place with the Kawa model improve well-being in community-dwelling older adults?

WELL-BEING, OCCUPATIONAL THERAPY, AND AGING IN PLACE

The focus of this pilot study was the psychosocial well-being of community-dwelling older adults. For the purposes of this study, two definitions of well-being guided the study design, the intervention, and the measurement of outcomes. First, well-being is an established outcome in the American Occupational Therapy Association's (2014) *Occupational Therapy Practice Framework* (3rd ed.), defined in part as "contentment with one's health, self-esteem, sense of belonging, security, and opportunities for self-determination, meaning, roles, and helping others" (p. S35). The goal of this study was to gain an understanding of each participant's perspective and experience of meaning, roles, and occupations as well as to provide informational resources that would create the means and opportunity for participants to actively determine their own future housing and community choices. Aging in place, which is promoted through this study, allows people to experience continuity of belonging, security, and roles without disruptive relocation.

Second, a useful conceptualization envisions well-being as the balance of an individual's physical, social, and psychological resources and the physical, social, and psychological challenges that he or she experiences at a given time (Dodge et al., 2012). The aging process, particularly during older adulthood, includes inevitable changes and challenges in physical health and functioning, role performance, social supports, and independence. These real or potential changes and losses can decrease well-being by upsetting the balance between a person's actual or perceived resources and challenges. Thus, this reimagining of psychosocial well-being was highly applicable in the context of an occupational therapy intervention to address issues of aging and aging in place. Essentially, the goal of this intervention was to provide additional resources for participants to counteract the aging-related challenges they face, either currently or in the future.

Occupational therapists are uniquely suited to address concerns about aging in place across the spectrum of human

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experience. Occupational therapy includes assessment and consideration not only of physical environments and functional performance of occupations but also of psychosocial considerations, such as well-being, life satisfaction, and quality of life (American Occupational Therapy Association, 2014). Occupation-based strategies for aging in place are not limited to the physical health, safety, and comfort of the client; they include temporal, personal, cultural, and virtual contexts as well (American Occupational Therapy Association, 2014). The occupational therapy process is centered on the client's ability to participate in chosen, necessary, and meaningful occupations; to preserve valued roles, habits, and routines; and to find satisfaction, enjoy quality of life, and ensure well-being throughout the life span (American Occupational Therapy Association, 2014).

Likewise, the *Occupational Therapy Practice Framework* (3rd ed.) affirms interventions designed for health promotion, modification/compensation/adaptation, maintenance of occupational performance, and prevention of disability (American Occupational Therapy Association, 2014). Education and training designed to support aging in place involve each of these approaches. Thus, the professional training and expertise of occupational therapists uniquely qualify them to assess the demands and effect of a multifaceted environmental context on a client's current and future ability to participate in chosen and necessary occupations (American Occupational Therapy Association, 2014, 2016; Kaminsky, 2010). Holistically addressing both current and future needs effectively places occupational therapists in a leading role among health care professionals pursuing the quadruple aim of health care.

THE KAWA MODEL

We chose the Kawa model of occupational therapy practice for use in this study as both a qualitative assessment tool and a means of designing and individualizing the intervention. The Kawa model was first conceptualized in 1999 by a group of Japanese occupational therapists and Japanese-Canadian scholar Michael Iwama (Murthi, 2016). The goals are to provide a culturally flexible model to aid occupational therapists to improve communication with clients, to better understand what a client finds meaningful and important, and to design optimal client-centered interventions (Teoh & Iwama, 2015). The model uses the metaphor of a river and its elements (water, rocks, driftwood, river walls and bottom, and the spaces between elements) to represent a client's life (Iwama et al., 2016). The Kawa model was designed to have multifaceted application within the context of the occupational therapy process, including use as "a conceptual model of practice, a frame of reference, assessment tool and modality" (Teoh & Iwama, 2015, p. 2).

An occupational therapy assessment or intervention that uses the Kawa model most often involves the client drawing a river that represents his or her life. The client draws elements, such as rocks and driftwood, to represent personal obstacles and attributes. The river walls and bottom, drawn in cross-section, represent the client's environmental context. Spaces between obstacles are areas of opportunity for growth, adaptation, and intervention to improve overall life "flow." The therapist asks open-ended, clarifying questions, using a collaborative approach to ensure that the model provides an accurate representation of the client's perceptions of his or her life. However, the model is designed to be flexible in application, and it can be used differently with each client, centering on the client's perspective and narrative rather than a specific procedural agenda (Teoh & Iwama, 2015).

Although no studies on the Kawa model have specifically addressed its use to support aging in place among U.S. community-dwelling older adults, earlier findings are consistent as to the benefits of the use of this model with varied populations. The Kawa model has been beneficial for facilitating communication, improving interpersonal relationships, and addressing problems as well as for team building with rehabilitation professionals (Lape et al., 2019; Lape & Scaife, 2017). Paxson et al. (2012) found that the use of the Kawa model improved client-therapist engagement, collaboration, and communication. In interventions with mental health populations, the Kawa model has increased client-client and therapist-client alliances and aided in destigmatization (Dellow & Skeels, 2016; Janus, 2017). Studies have assessed the use of the Kawa model as a qualitative assessment/reassessment tool and noted its efficacy in identifying clients' strengths and barriers and improving communication and collaboration (Carmody et al., 2007; Gregg et al., 2015; Leadley, 2015; Yeh et al., 2016). Further, the Kawa model allows valuable information to be obtained in a relatively short time frame (Janus, 2017).

However, the model is not without criticism. According to Wada (2011), the level of ambiguity in the model and its elements could limit accurate descriptions of occupational meaning, and the model gives insufficient attention to the discrete inner self of the individual (the idea of the self as unique, autonomous, and separate from his or her environment) as well as the concept and occupation of belonging

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(the roles and routines associated with active participation and belonging to a social group). Others have noted that therapists' discomfort or inexperience with the model and the imposition of therapists' own views or biases may limit its effectiveness (Carmody et al., 2007; Paxson et al., 2012). Practitioners also must understand both the foundational concepts of the Kawa model and basic psychology to apply the model effectively (Janus, 2017).

The pilot study described here was community based rather than home based, with all assessment and intervention completed at a YMCA facility rather than in clients' homes. Because aging in place directly relates to the client's physical home environment, it was vital to ensure a good understanding of each participant's home setup, environment, and physical needs as well as socioeconomic and relational contexts and perceptions. We hypothesized that the elements of the Kawa model would be useful for describing all of these aspects of aging in place.

Perhaps most important to this study, the metaphor of a river as representing a client's life has a direct correlation to well-being, the primary outcome being assessed. The model has been described as specifically representing client well-being: a client river that shows a "strong, deep, unimpeded flow" represents optimal well-being (Iwama et al., 2009, p. 1129). In this way, the Kawa model provides an explicit means to assess client well-being qualitatively and uses a collaborative approach to aid the client in assigning meaning and value to various elements of the environment (Iwama et al., 2009). These considerations are essential when considering how to design and implement proactive, client-centered strategies for improving readiness to age in place that are both feasible and meaningful for the client.

METHODS

For the pilot study, we chose a pretest-posttest mixed methods design to promote a comprehensive and holistic exploration of the problem, outcome, and lived experiences of the individual participants (Creswell, 2009). Convenience and snowball sampling methods were used for recruitment. All participants provided informed consent. The pilot study was approved by the institutional review board at Chatham University.

Study Participants

Participants were required to be 65 years or older, interested in aging in place, and able to provide or arrange for their own transportation. Membership in the YMCA was required because the intervention took place under the umbrella of YMCA programming and within YMCA facilities. Additionally, participants were required to be oriented to person, place, and time, and to give informed consent. Formal cognitive tests were not administered; it was assumed that participants who were able to meet the study criteria had the functional cognition required to participate in and benefit from the program. Seven older adults participated in the study. All were White and 69 to 92 years old. **Table 1** shows their demographic features.

Intervention

The intervention consisted of one-on-one education and instruction provided by the first author (an occupational therapist). One married couple completed the intervention together, but each spouse completed the quantitative measures independently. During the first session, each participant completed two quantitative surveys and participated in the Kawa activity. The first author (R.S.N.) demonstrated the use of the Kawa model by drawing her own river, including true-to-life examples of personal challenges (e.g., a cross-country move) and supports (e.g., supportive family). This therapeutic use of self was intended to build rapport and ease any hesitation of the participants to share personal stories and challenges (Taylor, 2008). Participants were then given the option of drawing or having the first author draw their own river model. This practice is not common, but is supported in the literature (Gregg et al., 2015; Iwama, 2006) and was intended to ease any discomfort or self-consciousness of the participants with use of an unfamiliar medium for storytelling as well as to introduce the concepts and river elements clearly. Each participant then described his or her own life related to aging in place, including occupations, environmental barriers and supports (e.g., presence of stairs vs. a flat entry to the home), health-related concerns, and social and community supports for aging in place. The first author drew each element, provided gentle prompting, and used closed- and open-ended questions to elicit detailed descriptions and perceptions. Informal member checking was used throughout the process to ensure that the features of the river were sized and placed accurately and were interacting in accordance with the participant's perspective. In this way, the first author made every attempt to act as "simply a guide to facilitate the service user's story telling" (Leadley, 2015, p. 50).

The first author then used the information and insights gained during the Kawa activity to design a participant-

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TABLE 1

Participant Demographics

Participant	Sex	Age, years	Marital status	Homeowner	Previous home modifications or caregiving in the home	Activity/independence level
A	Male	75-79	Widowed, living alone	Yes	Yes (spouse required one floor, wheelchair-accessible setup)	Fully independent. Drives. Formally retired but very active in volunteer work.
B	Male	86+	Widowed, living alone	Yes	No	Active member of YMCA; fully independent; employs housekeeper. Drives.
C	Male	86+	Divorced, living alone	Yes	No	Active member of YMCA and in the community; fully independent; employs housekeeper. Drives. Completes most yard/garden work independently.
D	Female	70-74	Married, living with spouse	Yes	Yes (spouse required modified setup after surgery/complications)	Retired but fully independent and active. Volunteers regularly with church and community organizations. Drives.
E	Female	65-69	Married, living with spouse (participant F)	Yes	Yes (had been primary caregiver for mother-in-law with functional limitations)	Retired. Active and independent; no limitations. Volunteers regularly in the community. Drives.
F	Male	70-74	Married, living with spouse (participant E)	Yes	Yes (mother; as noted for participant E)	Retired but still works as needed (truck driver). Volunteers in the community. Fully independent.
G	Male	80-85	Married; living alone (spouse recently relocated to long-term residential care facility)	Yes	Yes (cared for spouse with dementia until her placement in long-term residential care facility)	Retired. Active with YMCA, church, community. Drives. Visits spouse in residential care facility almost daily.

specific educational intervention during subsequent one-on-one sessions. The intervention included didactic teaching, discussion, review of handouts and printed resources, and demonstration of how to access online resources. The first author also drafted a written action plan and a to-do list for each participant for implementing strategies for aging in place. The educational subject matter, action plan items, and number of educational sessions (between two and five) varied based on participant needs and schedules. Subjects addressed with various participants included home modifications, fall prevention strategies, general health and wellness strategies, sleep and rest, community resources, tips for living with low vision, medication management, technologies for aging in place, strategies for adjusting to changing roles and maintaining meaningful occupation, and support groups in the community.

In each participant's final session, the first author and the participant collaborated to finalize the individualized action plan for implementing the aging in place strategies that were discussed in earlier sessions. Final questions and concerns were addressed as needed, with the action plan adjusted collaboratively. The quantitative surveys were readministered, and the Kawa activity was repeated as a

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qualitative assessment of postintervention perceptions of well-being.

Outcome Measures

As described previously, the Kawa model was used as a qualitative assessment. Although it is relatively untested among U.S. older adults, this type of before-and-after comparison of a client's Kawa model has been noted to be an effective qualitative assessment in other populations (Carmody et al., 2007; Gregg et al., 2015; Leadley, 2015) and is affirmed by Kawa scholars (Teoh & Iwama, 2015).

To provide quantitative data to complement the Kawa model and triangulate the results, two Likert-type scales were used to measure well-being. The Psychological General Well-Being Index (PGWBI) is a 22-item scale that measures overall psychological well-being. Test items are grouped into six subcategories (anxiety, depressed mood, positive well-being, self-control, general health, and vitality). The test has been shown to have adequate reliability and validity among various populations, including older adults (Narushima et al., 2013; Peto et al., 2001). In addition, the first author developed a Likert-type survey to measure well-being specifically as it relates to aging in place. This survey contains 14 items, each relating specifically to at least one aspect of well-being, as described by the definitions noted earlier. Each item also relates to perceived ability and preparedness to age in place. Before study implementation, this survey was reviewed by 10 occupational therapists for content validity and piloted with four older adults who were not involved with the project. Based on feedback, the measure was revised for clarity and comprehensiveness. The preintervention survey also included several demographic questions. The survey included a total of 14 items (excluding demographic questions) that participants rated on a scale of 1 to 5, from "strongly agree" to "strongly disagree." Participants scored items as 5 to indicate the highest sense of well-being and as 1 to indicate the lowest. The items are shown in **Table A** (available in the online version of the article).

Data Analysis

Member checking and note taking were used throughout the Kawa activity to ensure that the drawn representation of the client's views was accurate as expressed in the river drawing. After they completed the second Kawa activity, the first author and the participant collaboratively compared the pre- and postintervention rivers. Specifically assessed were the presence or absence and relative size of obstacles (rocks) and assets (driftwood), the quality of the environment (width and smoothness of the river walls and bottom), and the depth and flow of the water. They continued the discussion until the participant stated that his or her views were accurately represented and understood. The first author took notes during and immediately after the discussion and transcribed her notes to digital form. Thematic analysis, as outlined by Braun and Clarke (2006), was used throughout the process to assess for qualitative changes and themes. The second author (J.E.L.) reviewed the transcribed notes and Kawa drawings separately and completed an independent thematic analysis. The two authors then met to discuss the themes and found no discrepancies, despite some variation in terminology used to describe the overarching concepts. This process helped to ensure consistency and lend trustworthiness to the data analysis.

The PGWBI was scored for each participant according to the official PGWBI scoring guide (Dupuy, 1990). Descriptive statistics were used to calculate the mean, median, and (when available) mode of the aggregate participant scores, and the results were examined to identify patterns and change over time. The second author reviewed the data and statistical procedures for accuracy. Further inferences from the data were limited because of the pilot nature of this study, the small homogenous sample size, and the short duration of the study. Inferential statistics were not used because the goal of this study was to explore the use of the Kawa model as a tool to improve well-being by providing education on aging in place, and any improvement in well-being was viewed as clinically significant.

RESULTS

The use of the Kawa model for assessment provided rich qualitative data and insights on perceptions of current and anticipated supports and barriers to aging in place. Most participants started at generally high levels of overall well-being, as shown by relatively few rocks (barriers or obstacles), relatively smooth river walls and bottom (physical and social environment), and substantial amounts of driftwood (assets and personal strengths). **Figure 1** shows a sample of participants' Kawa river models.

Several themes emerged during the Kawa activity that indicated that perceptions and/or well-being had improved. These themes were developed and documented throughout the study (Braun & Clarke, 2006; Creswell, 2009). General qualitative themes included increased peace of mind, immediate action to implement plans to

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prepare for aging in place, affirmation of preparations already made, and positive effects on family/support persons. The themes are outlined in **Table B** (available in the online version of the article).

For both quantitative measures, a higher score indicated greater well-being (Dupuy, 1990). For the PGWBI, five of seven participants showed increased global scores, as did the group in aggregate (**Figure 2**). For the five positive scores, improvement ranged from 0.9 point to 12.7 points (+0.09% to +12.7%). The two negative scores were -3.6 points and -0.7 points (-3.6% and -0.07%, respectively). All participants showed improvement in the positive well-being subsection of the PGWBI, with a mean improvement of 7.8% for the cohort at large.

For the author-generated survey, six of seven participants showed improvement in well-being, as did the group in aggregate. The mean for the group was an improvement of 3.4 points (4.9%). The largest positive point spread was 9 points (12%), and the smallest was 2 points (3%). One participant had a negative point spread of 2 points (3% regression) from pre- to postintervention (**Figure 3**).

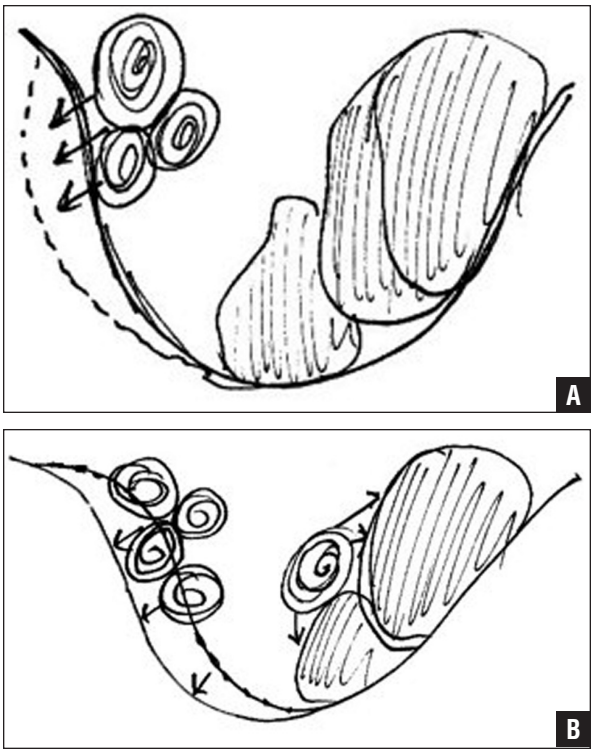

Figure 1. Sample Kawa model drawing in cross-section, pre- and postintervention. (A) Preintervention: Large rocks blocking the flow on the right side represent the participant's grief at the recent loss of his spouse, health concerns, the taxing nature of volunteer responsibilities, and uncertainty about planning for future needs. Driftwood (left) represents family/social support, an amenable home setup, and fulfillment in volunteer duties. The solid line represents the preintervention environment, which is relatively smooth and shows driftwood applying force (arrows) to represent the participant's goal to use driftwood to make space (dotted line) to improve flow. (B) Postintervention: Additional driftwood on the left represents the participant's new knowledge about resources for aging in place and his decision to reduce volunteer duties to regain occupational balance. On the right, driftwood shown applying force (arrows) to rocks represents a specific plan to seek grief counseling because during the activity the participant recognized that his unaddressed grief was hindering his well-being. Postintervention, the participant deliberately omitted one large rock, representing the removal of uncertainty about aging in place. The solid line on the left represents the participant's sense of increased space, with arrows again representing driftwood applying force to bypass or remove barriers.

DISCUSSION

Well-being

Both the qualitative and quantitative data show that participants' overall well-being increased through provision of education on the strategies and available resources to support aging in place. This increase is shown by modest but consistent improvement in aggregate well-being scores in both the PGWBI and the author-generated well-being survey as well as in the themes that emerged during the Kawa activity. Only one participant's (Participant B) scores indicated a decrease in well-being between pre- and postintervention on both surveys; however, this participant was one of the most vocal and enthusiastic in expressing appreciation of the program's benefits. It is possible that the measures may have highlighted the small fluctuations in well-being that humans experience daily. It is also possible that the participant interpreted items on the PGWBI and the author-generated survey differently when taking the surveys for the second time.

Statistical inferences were not possible because of the small sample size, the short duration of the study, and the relatively high levels of preintervention well-being. However, the quantitative results are considered clinically significant because even modest changes in well-being can contribute to quality of life and may be a protective factor for health maintenance (Steptoe et al., 2015). The participants had short-term improvement in well-being and were given tools and education to improve their ability to age in place. This pilot study may shape future programming choices at this YMCA and could have further reach if expanded and implemented as a permanent program or used as a guide for similar programs elsewhere.

Use of the Kawa Model

The participants were unfamiliar with the Kawa model before the study. Some initially expressed skepticism about the use of the metaphor when it was explained. However,

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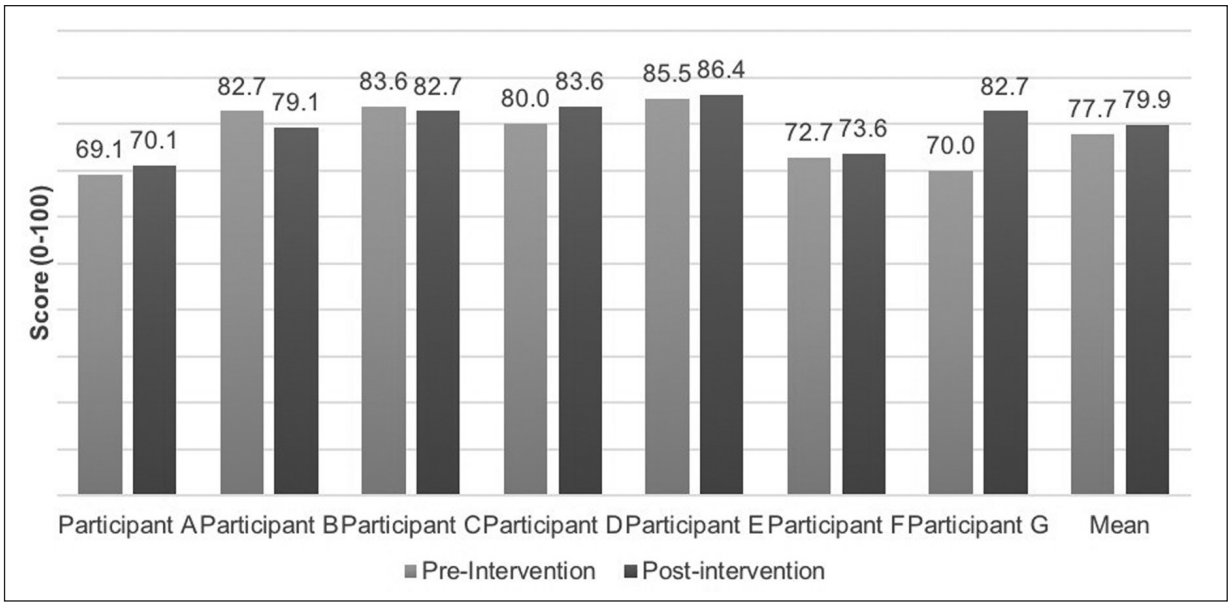

Figure 2. Pre- and postintervention global scores for the Psychological General Well-Being Index, by participant. Higher value indicates a positive result.

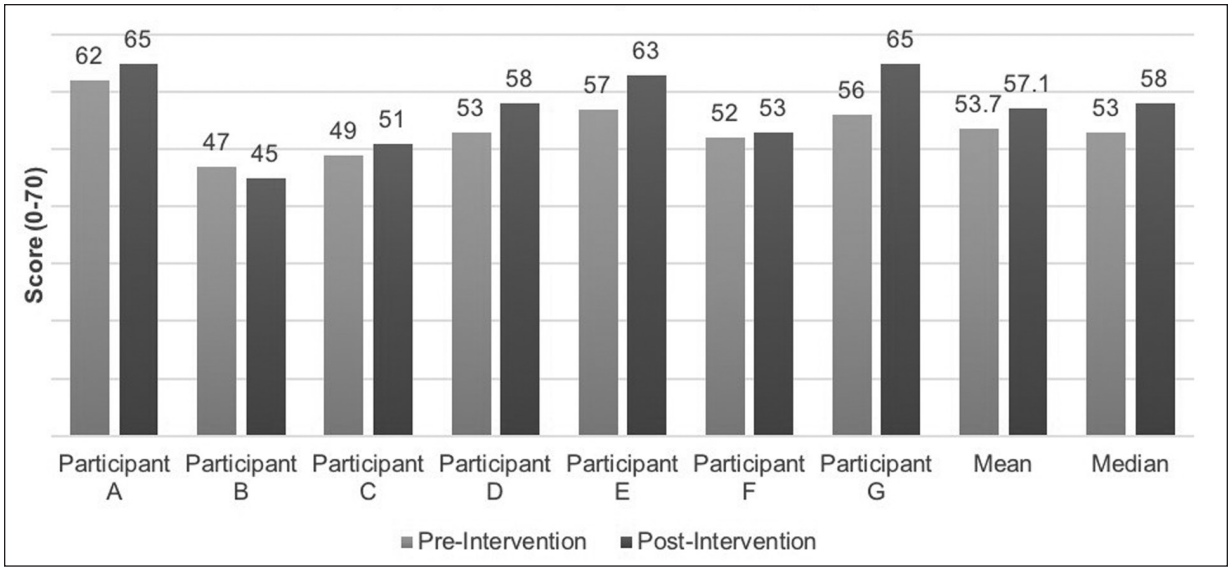

Figure 3. Pre- and postintervention scores on the author-generated survey of well-being, by participant. Higher value indicates a positive result.

once the activity was underway, all participants actively engaged in the process. Particularly during the postintervention Kawa activity, all participants showed familiarity and competence with identifying and characterizing the elements and readily offered corrections or modifications to the first author's initial drawing (e.g., requesting that items be drawn larger, smaller, or in a different location, and offering pertinent explanations for changes). **Figure 1** shows a sample of pre- and postintervention drawings.

The use of the Kawa model during the intervention greatly enhanced communication and rapport, facilitated client engagement, and provided the first author with insight as to each individual's needs, values, roles, challenges, and supports. These findings align with the documented benefits of this model in various settings (Carmody et al., 2007; Gregg et al., 2015; Lape et al., 2019; Lape & Scaife, 2017; Leadley, 2015; Paxson et al., 2012). For some participants, the river metaphor seemed to serve as

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an especially useful projection; in describing obstacles in the more neutral terms of the metaphor, participants may have been more willing to offer details than they would have been in a simple question-and-answer interview. This finding was evidenced by the initial use of more general descriptions (e.g., identifying "volunteering" as driftwood, indicating a valued social occupation) that were refined according to the terminology of the river elements (e.g., suggesting that the driftwood should be stuck on a rock because of current challenges with volunteer duties). Additionally, the drawing or river construction aspect of the Kawa model may support a more collaborative, relaxed interaction specifically because it does not reflect a traditional clinical client-provider interaction. Parallels can be drawn between these observations and those noted by both Paxson et al. (2012) and Leadley (2015) in their respective applications of this model with mental health populations.

Overall, these results suggest that the Kawa model may be highly effective when used as an assessment and a design tool for this type of intervention. Additionally, the results of this pilot study support the existing literature showing that a proactive, client-centered, individualized approach to interventions centering on wellness, productive aging, and aging-related issues can improve current well-being and future success in aging in place (Arbesman & Lieberman, 2012; Clark et al., 2012; Graham et al., 2014; Rose et al., 2010). Although much of the literature focuses on more general wellness and productive aging strategies for this population, this study provides early evidence that educational interventions specific to community-dwelling adults seeking to age in place may improve well-being and that the Kawa model may be effective for guiding and assessing interventions.

Additionally, the more relaxed nontraditional setting and enhanced client-practitioner interactions align with the quadruple aim of health care. The preventive or proactive nature of the intervention generally leads to improved outcomes and cost-effectiveness (Arbesman & Lieberman, 2012; Clark et al., 2012). Enhanced communication, rapport, and individualization of treatment improve clients' experience of care (Kuipers et al., 2019; Poey et al., 2017) and increase providers' satisfaction with delivering care, which can help to prevent burnout and disengagement (Aiken et al., 2012; Chang et al., 2009).

This pilot program was conducted on a volunteer basis to explore the use of this model and this type of programming in a community setting. Given the innovative and nontraditional setting of this study, to provide reimbursement for similar programs in the future, creative means of funding would need to be explored. Some occupational therapists are beginning to enter into mutually beneficial partnerships with senior housing organizations, senior centers, builders, and advocacy organizations for the aging. This type of Kawa-based education for older adults seeking to age in place would fit well with this approach and would move occupational therapy into the forefront of preventive, community-based care, which supports each aspect of the quadruple aim.

Limitations

The time frame between pre- and postintervention administration of the PGWBI may limit its sensitivity to change. During this pilot study, postintervention outcome measures were obtained 2 to 10 days after the initial assessments, rather than at the recommended 1-month time frame (Dupuy, 1990). To maximize participation, sessions were scheduled at the convenience of the participants, which limited later follow-up. Additionally, although the author-generated survey was piloted with appropriate populations and reviewed for content validity by occupational therapists, it has not been formally tested. Participants were recruited from only one site. The sample size was small ($N = 7$) and relatively homogenous, which limits generalizability. The first author was the only person who took notes during each individualized session, which could lead to investigator bias, although member checking was used throughout to verify the perspectives of the participants. The effectiveness of the intervention may be limited if it is administered by a practitioner who is inexperienced with the Kawa model, as noted by Carmody et al. (2007) and Paxson et al. (2012). Although the first author was familiar with the Kawa model, this was the first use of the model with this specific subject matter and population. The second author has extensive experience in clinical application and research with the Kawa model in various contexts (e.g., Lape et al., 2019; Lape & Scaife, 2017) and provided mentoring for the first author throughout the study. In addition, another colleague provided peer debriefing during thematic analysis. However, the effect of inexperience cannot be excluded. Future studies should include larger, more diverse samples and the opportunity for long-term follow-up to allow further inferences from the data.

CONCLUSION

Despite the limitations noted, the findings from this pilot study support the existing literature on the Kawa

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model and the use of a proactive, preventive approach to aging-related education. These findings align with and support the goals of the quadruple aim of health care and affirm the specific value of occupational therapy in addressing concerns about aging in place. Further studies are needed to add momentum to the shift toward preventive, community-based, client-centered occupational therapy.

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Table A. Survey

1. I feel content with my current state of health.
2. I feel a sense of belonging in my home and community.
3. I have no concerns about staying in my current home throughout my life.
4. I am able to assist other members of my household as needed or arrange for others to assist them.
5. I worry about knowing how to care for myself and/or my spouse at home in the coming years.
6. I have strong social support that helps me meet the demands of life.
7. I am satisfied with my ability to fulfill the expectations of my friends and family.
8. I worry about falling in my home.
9. I don’t know if I need to make changes in my home to stay in it as long as possible.
10. I feel capable of determining my own future living arrangements.
11. I feel confident that I know what resources are available to me as I age.
12. I feel safe and secure in my home environment.
13. I have concerns about how to keep living a meaningful life as I age.
14. I worry about whether I will have the physical, social, and/or financial resources I need to meet the demands of aging in my home environment.

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Table B

Qualitative Themes That Emerged During the Kawa Activity.

Theme		Quotes/Examples
Preparation/plan-making led to increased peace of mind, relief of worries, ability to envision change in a positive way.		"This just makes me feel better to do this" (B) "Having a plan helps" (D) "I feel more relieved" (A) Obstacle (rock) representing stairs was removed once given information re: stair glide. (B) Addition of driftwood representing knowledge to remove rocks/barriers. (C) Addition of driftwood representing cognitive processing of grief during the educational sessions and subsequent days; rock representing grieving process was smaller during post-intervention Kawa. (G)
Organized plan and goals refined during the post-intervention Kawa activity will help to generate action.		"This will help me actually do this stuff rather than just thinking about it" (C) "If I don't have a plan, I just put things aside" (C) "It helps to have things organized like this" (D)
Addition of resources (driftwood) did not always remove barriers (rocks) but increased space or opportunity for life flow (water).		Participant A: Addition of driftwood representing plan/intent to seek grief counseling to regain occupational balance. Participant D: Second river model purposefully drawn deeper to represent new knowledge and plan making space for improved flow.
New knowledge and identification of rocks (barriers) led to immediate action to remove barriers.		"I already called [referred resource] since I saw you (first author) last!" (A) "I got those railings installed right after you told me to" (B) "I was looking at the grab bars like you said, to see if they are installed ok" (F)
Family supportive of participation in the program and aging in place preparation; indicates that the intervention may also improve family/caregiver well-being.		"My son is thrilled I'm doing this" (A) "I told my family about this and they said it's great" (B) "Yeah, my grandkids are always getting on me about [potential fall hazard]. They'll be glad to know we talked about this" (C)

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The intervention and discussion during the Kawa activity served to affirm and encourage preparations already made by participants.		"…. Glad to know we're doing the right things" (E) "Good to have an expert agree with me (that) we set things up right." (G)

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